

Peer Review and the Diffusion of Ideas

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Abstract

This study examines a fundamental yet overlooked function of peer review: its role in exposing reviewers to new and unexpected ideas. Leveraging a natural experiment involving over half a million peer review invitations covering both accepted and rejected manuscripts, and integrating high-scale bibliographic and editorial records for 37,279 submitting authors, we find that exposure to a manuscript's core ideas significantly influences the future referencing behavior and knowledge of reviewer invitees who decline the review invite. Specifically, declining reviewer invitees who could view concise summaries of the manuscript's core ideas not only increase their citations to the manuscript itself but also demonstrate expanded breadth, depth, diversity, and prominence of citations to the submitting author's broader body of work. Overall, these results suggest peer review substantially influences the spread of scientific knowledge. Ironically, while the massive scale of peer review, entailing millions of reviews annually, often drives policy debates about its costs and burdens, our findings demonstrate that precisely because of this scale, peer review serves as a powerful yet previously unrecognized engine for idea diffusion, which is central to scientific advances and scholarly communication.

Introduction

Peer review is a cornerstone of scientific practice, but its origins reveal an often-overlooked function beyond the familiar roles of providing feedback and quality control. Originally formalized in the 19th century by journals such as *Philosophical Transactions*, peer review was introduced to entice scientists away from informal socialization of their discoveries and towards formal dissemination through scientific journals [1]. The promise was simple but powerful: by submitting work to a journal, scientists were assured a readership—at least the handful of peers who would review their submissions. Thus, peer review emerged not merely as science’s gatekeeping tool, but fundamentally as an engine of knowledge diffusion, promoting the spread and adoption of new ideas.

Over the ensuing centuries, peer review’s foundational role as a diffusion mechanism has been overshadowed by its critical function in maintaining research integrity and quality [2-4]. Today, individual scientists collectively spend about 100 million hours per year on peer review [5], roughly equivalent to the time it would take to write millions of grant proposals [6], yet the system’s efficiency and effectiveness have remained subjects of ongoing debate and improvement efforts [7-9].

Despite the extensive focus on its evaluative role, little attention has been paid to peer review’s foundational role as a mechanism of diffusing new ideas, despite the fact that exposure and access to new ideas support innovation in science as much as the filtering process [10-12]. Yet, even as the overall volume of scientific knowledge has grown rapidly [13, 14], knowledge search progressively emphasizes a small fraction of work published in the leading journals, informatics search tools present knowledge by popularity, and lengthening journal publication lags slow access to new knowledge [13, 15, 16]. Within this context, peer review invitations represent unique micro-exposures to emerging research, potentially counteracting trends towards narrowing scholarly attention by serendipitously exposing reviewer invitees to new ideas.

Here, we quantify peer review’s original promise as a conduit of knowledge diffusion. For example, does early exposure to research work during peer review promote the adoption of new ideas among reviewer invitees? Do the characteristics of reviewer invitees’ citations to work they

are exposed to in the review invitations expand the breadth, depth, diversity, and prominence of citations, enriching science and spreading peer review costs over more benefits? These questions have remained elusive due to substantial empirical challenges. Such research requires comprehensive, longitudinal data on editorial decisions for both accepted and rejected manuscripts, detailed professional identities of authors and reviewer invitees, citation behaviors, and insights into the nuances of editorial processes—data typically restricted or inaccessible, partly due to the inherent anonymity and confidentiality that underpins peer review.

Here, we conduct one of the first studies of links between peer review and the diffusion of new and related ideas. We use a unique 10-year, longitudinal dataset that combines 551K peer review invitations from a leading medical journal with detailed bibliometric records on the professional identities of authors and reviewer invitees. We then merge this data with SciSciNet [17] and Web of Science (WoS), large-scale publication and bibliographic databases. Together, the data allow us to identify the reviewer invitees’ history of research, expertise, response to the review, and what they later cite and publish, including new and related work by the submitting author. This rare combination of editorial records and bibliographic metadata enables the first systematic test of whether peer review is related to the spread of scientific ideas.

We quantify a reviewer invitees’ adoption of new and related ideas into their research by examining their citation behavior toward the submitting author’s work before and after exposure to the review invitations, employing a difference-in-differences (DiD) regression framework. In addition to overall citation frequency, we quantify citation quality through measures of prominence, as well as citation breadth, depth, and diversity.

A key analytical challenge in quantifying peer review’s diffusion effect lies in reviewer self-selection. Comparing individuals who actively reviewed manuscripts with those who did not can be problematic because reviewers typically choose manuscripts that align closely with their interests, expertise, and prior knowledge, thus introducing significant selection bias. Our methodological innovation addresses this challenge by focusing exclusively on individuals who did not conduct the review. Specifically, we compare two distinct groups of invited reviewers: those who actively decline the invitation by clicking a link (Click-Denial) and those passively

declined by the journal's automated system (Auto-Delay). While these two groups are comparable in that neither group engages fully with the manuscript, they differ in that the Click-Delay reviewer invitees likely viewed essential manuscript metadata, such as title, abstract, and authorship before declining, whereas Auto-Delay reviewer invitees likely did not. This approach allows us to isolate incidental exposure to a manuscript's core ideas independent of self-selection. To strengthen our assumption about differences in Click-Delay and Auto-Delay groups' likely exposure to a manuscript's core ideas, we further exploit a natural experiment: the quasi-random timing of peer review invitations—sent on either a weekday or weekend—which affects the likelihood of invitee engagement independently of their research interests or citation intentions. [18].

Across diverse tests and robustness checks, we consistently find that concise exposure through peer review invitations meaningfully promotes the dissemination of new ideas. These findings demonstrate that peer review functions not only as a gatekeeping institution but also as a subtle yet powerful engine of knowledge diffusion – an essential and previously overlooked benefit embedded within this foundational institution of science.

Data

We analyzed the complete peer review records of one journal from a premier publisher of medical research over a ten-year span (2012-2022), encompassing 37K submissions and 551K review records¹. Each record captures the full invitation process, including metadata about the manuscript—title, abstract, author names, and the corresponding author—as well as timestamped reviewer invitees' responses (Figure 1A). The data were collected by the journal through their ScholarOne intake platform (formerly Manuscript Central). Besides the fine-grained data collected by ScholarOne, a unique feature of these data is that ScholarOne's broad use across science standardizes scholarly editorial workflows and peer review processes despite journal-specific or author-specific characteristics of different fields. Over 9,000 diverse journals and top publishers use ScholarOne (www.silverchair.com) to manage over three million manuscripts annually as part of their peer review and submission processes. Thus, while peer review's filtering role can be moderated by journal-specific and author-specific characteristics [4],

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ScholarOne employs the same standardized intake and review system similarly across journals, manuscripts, and reviewer invitees.

Upon receiving an invitation to review, reviewer invitees may respond in one of three ways (Figure 1B): (1) Accept—clicking to accept the invitation; (2) Click-Denial—explicitly declining by clicking “Declined” or “Unavailable”; or (3) Auto-Denial—not responding to the invitation, with the system logging it as a passive decline.

This setting offers a rare window into an otherwise invisible part of scientific communication: micro-exposures to emerging scientific content—delivered through manuscript titles, abstracts, and author lists—well before publication. The data’s fine-grained structure allows us to observe and differentiate these exposures at scale.

We then link this dataset to SciSciNet [17] and WoS, allowing us to track individual-level bibliometric records and the subsequent citation behaviors of each reviewer invitee over time (See details in SI S1). For every reviewer invitee, we observe both the manuscripts they were invited to evaluate and their longitudinal publication and citation trajectories.

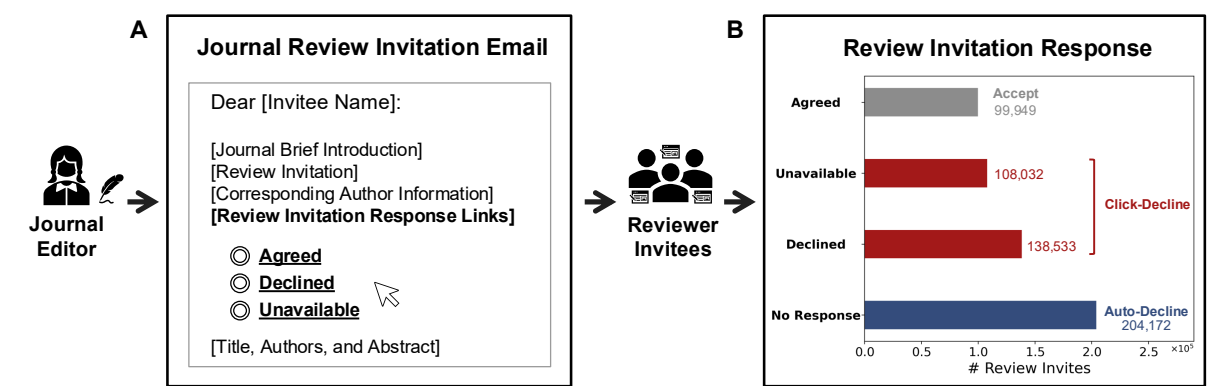


Figure 1. The peer review invitation process and classification of reviewer invitees’ responses. (A) Editors send email invitations containing metadata about the journal and manuscript, along with embedded response links: “Agreed,” “Declined,” and “Unavailable.” (B) We group reviewer invitees based on their responses to the invitation: the Click-Denial group includes 247K invitees (44.77%) who actively decline the review request; the Auto-Denial group consists of 204K (37.08%) invitees who passively non-respond to the review request; and the Accept group has 100K individuals (18.15%) who actively accept the review.

Results